PMO PORTFOLIO ANALYSIS

Data Analyst Case Study

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Tools: Excel, Power Query, Power BI and DAX

Executive Summary

This case study looks at how a Project Management Office (PMO) can review a portfolio of projects and decide where management attention may be needed. The provided workbook contains 1,696 monthly project records for 100 projects, covering delivery, cost, risk, resources and benefits. I explored the workbook, checked key data relationships, built a Power BI model and created a four-page report. Rule-based checks classify projects into actions such as Recover, Pause for Review, Accelerate and Review Data Freshness. These are analysis-based suggestions, not final management decisions.

Portfolio action	Projects	Share
No Further Intervention	43	43%
Recover	29	29%
Review Data Freshness	18	18%
Accelerate	6	6%
Pause for Review	4	4%
Monitor	0	0%

The counts total 100 projects. No Further Intervention includes Completed and Terminated projects; it does not mean all those projects were successful. Review Data Freshness is a data-quality action, not a delivery decision.

1. Domain Overview

A Project Management Office (PMO) helps a company track and govern its projects. Each project uses money, people and time. The PMO helps leaders see whether projects are progressing as planned, whether costs may exceed budgets, whether risks are increasing, whether teams have enough capacity and whether expected business benefits are likely to be delivered.

A project portfolio is the full set of projects being managed. A programme is a group of related projects. Project health combines signals such as schedule, cost, risk, resources and stakeholder confidence. Strategic alignment indicates how closely a project supports company goals. Benefits are the business value expected from a project, such as savings, better service or new capability.

Simple glossary

Term	Meaning
Baseline budget	The approved budget used as a comparison point.
Forecast total cost	The estimated final cost based on the available data.
Forecast budget overrun	Forecast total cost minus baseline budget. A positive value means forecast cost is above baseline.
Progress gap	Actual progress minus planned progress, shown in percentage points. A negative gap means behind plan.
Risk exposure score	A source-provided score used to compare project risk; its exact formula was not independently recreated.
Resource capacity gap	Available resource hours minus planned hours. A negative value indicates a shortage.
Benefit realisation	Benefits realised to date compared with benefits planned to be delivered by that date.
Snapshot freshness	Whether the project's latest available record matches the portfolio's latest reporting date.

2. Business Problem and Objectives

Management needs a clear view of which projects are performing well and which may need help. Without a joined-up view, teams may miss delivery delays, forecast overspending, resource shortages, high-risk projects or gaps in expected benefits. The dashboard brings these signals together to help leaders decide what to investigate first.

Main objective: use project data to support better portfolio review and prioritisation.

Key business questions

- How many projects are Active, Completed, Paused or Terminated?

- Which Active projects have the lowest health scores?
- Which projects are furthest behind plan or have the largest forecast overruns and delays?
- Which projects have high risk exposure, overdue mitigation actions or overdue milestones?
- Which business units have the largest resource shortages?
- Where are benefit shortfalls largest, and how much benefit has been realised?
- Which projects meet the agreed rules for recovery, pause or acceleration?
- Which project records are older than the portfolio's latest reporting date and should be verified?

Stakeholders

Intended users include PMO leaders, portfolio managers, project managers, finance teams, resource planners, risk teams and senior decision makers. They can use the report for delivery follow-up, financial control, risk review, capacity planning and portfolio prioritisation.

3. About the Data

The source was the provided Excel case-study workbook "PMO Portfolio_From Strategy to Success_C41.xlsx". It contains a monthly fact table, dimension tables and a data dictionary. One row in FactProjectMonthly represents one project at one reporting date.

Item	Details
Projects	100 unique projects
Monthly fact rows	1,696 rows in FactProjectMonthly
Grain	One project per reporting date; ProjectKey + ReportingDate uniqueness checked
Calendar range	DimDate covers 1 Jan 2023 to 31 Dec 2026
Project reporting range	October 2023 to September 2026
Dimensions	Project, Programme, Business Unit, Employee, Location, Delivery Model, Strategic Theme and Date
Key fields	Status, RAG, planned/actual progress, baseline/actual/forecast cost, forecast delay, risks, resource hours, benefits and health score
CurrentProjectSnapshot	Power BI calculated table with one latest available record per project

Data model and reporting views

The model follows a star-style structure. FactProjectMonthly stores repeated project performance records, while dimension tables provide descriptive fields for filtering and grouping. Relationships were created from dimension tables to the fact table and CurrentProjectSnapshot. The employee relationship uses ProjectManagerKey; SponsorKey would need a separate relationship if sponsor analysis is added.

FactProjectMonthly is used for monthly trends. CurrentProjectSnapshot is used for current portfolio comparisons because it keeps one latest available record per project. However, latest available per project does not mean all projects have records from the same date.

4. Approach and Workflow

Stage	What was done	Tool / status
Business understanding	Defined questions around delivery, cost, risk, resources, benefits and decisions.	Analysis — completed
Data exploration	Reviewed workbook structure, tables, columns, date coverage and data dictionary.	Excel — completed
Data quality checks	Checked project-date uniqueness, dimension key uniqueness, join matches and selected milestone/risk logic.	Power Query / model checks — completed
Data modelling	Created relationships and CurrentProjectSnapshot with one latest row per project.	Power BI — completed
Measures and rules	Created DAX measures for portfolio, delivery, cost, risk, resources, benefits and decisions.	DAX — completed
Dashboard	Built four report pages with cards, charts, tables, slicers and report-page tooltips.	Power BI — completed
Final sharing	Final checks, publishing and links remain to be completed.	In progress

5. Data Cleaning and Preparation

- Checked the grain of FactProjectMonthly and verified that each ProjectKey + ReportingDate combination is unique, reducing the risk of counting a project-month twice.
- Checked dimension keys for uniqueness and checked that fact-table keys matched the dimensions.
- Reviewed date coverage: the date dimension spans 2023–2026, while project reporting records span October 2023 to September 2026.
- Created CurrentProjectSnapshot to keep the latest available row for each project, so historical rows are not counted as separate current projects.

- Created Severe Warning Count and Material Warning Count columns using explicit thresholds for delay, health, CPI, risk exposure, stakeholder confidence, resource shortage, benefit realisation and critical risks.
- Created Recommended Action and Recommendation Reason columns. The rule order checks Completed/Terminated status, stale data, Paused status, multiple severe signals, material signals and acceleration eligibility.
- Reviewed earned-value candidate calculations. Candidate earned value aligned with the supplied cost variance and CPI to within about one euro. The schedule variance definition/sign convention differed, so the supplied value was not silently replaced.
- Kept missing benefit-realisation values when the percentage may not be meaningful. A missing value should not automatically be changed to 0% without checking the planned-benefit denominator.

6. Power BI Report

The report has four pages. Business Unit and Programme slicers support exploration. The design uses a consistent theme, short subtitles, ranked charts and tables.

Page	Purpose and main visuals
1. PMO Portfolio Overview	Total Projects, Active Projects, Average Portfolio Health Score and Projects with Outdated Data cards; status chart; monthly RAG trend; projects by Business Unit; ten weakest Active projects.
2. Delivery & Financial Performance	Cards for projects behind plan, projects with forecast delay, forecast overrun and overrun percentage; progress-gap, forecast-overrun and delay charts; delivery and cost table; report-page tooltips.
3. Risk, Resources & Benefits	Cards for critical risks, overdue mitigation actions, resource shortage and benefit shortfall; risk exposure and benefit shortfall charts; resource shortage by Business Unit; attention table; supporting tooltips.
4. Portfolio Decisions	Five action cards; Strategic Alignment Score vs Portfolio Health scatter chart; actions by Business Unit; management attention table; acceleration-candidates table.

Key measure definitions

Measure	Definition
Progress Gap (pp)	(Average Actual Progress % – Average Planned Progress %) × 100.
Forecast Budget Overrun (€)	Total Forecast Cost – Total Baseline Budget. Positive means forecast cost exceeds baseline.
Overrun % of Baseline	Forecast Budget Overrun ÷ Total Baseline Budget.
Total Resource Shortage (Hours)	Sum of shortage hours from negative capacity gaps, expressed as positive hours.
Benefit Shortfall to Date	Planned Benefits Due to Date – Benefits Realised to Date.
Snapshot Freshness	Stale when a project's latest available reporting date is earlier than the latest reporting date in the snapshot.

7. Analysis and Key Findings

These findings reflect the provided case-study workbook and the current report. They are not verified outcomes from a named company.

Question	Finding	Meaning
Portfolio size and status	100 projects. Latest available status per project: 55 Active, 40 Completed, 3 Terminated, 2 Paused.	Status is based on each project's latest available record, not one common reporting date.
Data freshness	61 projects have latest records older than the portfolio's latest reporting date, 30 Sep 2026.	Verify records before making current decisions. This is a relative freshness check, not a fixed 30-day age test.
Weakest Active projects	The weakest-project table ranks Active projects by PortfolioHealthScore. Several displayed projects have scores from 48 to 61 and show delays or negative resource gaps.	Prioritise the lowest-scoring Active projects, but check data freshness first.
Delivery issues	49 projects are behind plan and 48 have forecast delay under current measures and filters.	Review the most negative progress gaps and largest forecast delays.
Financial pressure	Net forecast budget overrun is about €12M, or 13.8% of baseline budget under the current report context.	Investigate largest overruns and confirm whether estimates or approved changes explain them.
Risk concerns	42 projects have critical risks and 107 overdue mitigation actions in the report context.	Review critical risks and overdue actions with project owners.
Resource shortage	Total resource shortage is about 34K hours. Risk & Compliance shows about 11.4K hours in the displayed chart.	Review resource allocation and capacity options in the most affected units.
Benefits	Benefit shortfall to date is about €34M; benefit realisation was about 72% under the earlier KPI calculation.	Review the largest shortfalls and check the planned-benefit denominator.
Rule-based actions	43 No Further Intervention; 29 Recover; 18 Review Data Freshness; 6 Accelerate; 4 Pause for Review; 0 Monitor.	29 projects need a recovery plan; 18 need data verification; six meet initial acceleration criteria.

RAG note: the monthly trend shows 34 Green and 5 Amber records in September 2026, with no Red records visible that month (39 records total). The latest-available-per-project snapshot has a different population and distribution. Do not mix these two views when describing portfolio health.

8. Recommendations

Action	Basis	Suggested next step
Review Data Freshness (18)	Latest record is older than 30 Sep 2026.	Ask project owners to confirm status, forecast, risks, resources and benefits before other decisions.
Recover (29)	At least one material warning signal is present, unless a higher-priority rule applies.	Create a recovery plan with owner, action, due date and target measure.
Pause for Review (4)	Paused status or three or more severe warning signals, after higher-priority rules.	Review dependencies, risk, cost and business need before authorising further work.
Accelerate (6)	Active, Green, health ≥80, delay ≤30 days, CPI ≥0.90, and benefit condition met where benefits are due.	Confirm strategic priority and capacity before approving acceleration.
No Further Intervention (43)	Completed or Terminated under the current rule order.	Keep out of active intervention queues; retain closure and financial records as needed.
Termination review	No automatic Terminate rule was implemented.	Do not terminate based on this dataset alone. Define thresholds and consider strategic, cost, risk, contract and benefit evidence.

These are case-study recommendations, not company policy or automatic approvals. Strategic Alignment Score can support prioritisation, but its meaning and cutoffs should be agreed before it is used for decisions.

9. Limitations and Next Steps

- The workbook is a provided case-study dataset; findings should not be presented as verified results from a named company.
- Project records are not all from the same date. CurrentProjectSnapshot keeps each project's latest available record, so stale records can affect comparisons.
- The outdated-data count uses a relative cutoff (earlier than 30 Sep 2026), not a fixed 30-day age test.
- Action categories depend on thresholds selected for this case study and should be reviewed with PMO leaders before real use.
- The RiskExposureScore formula was treated as source-provided; its exact construction was not independently recreated.
- BenefitRealisationPct can be blank when unavailable or not meaningful; investigate missing values rather than automatically replacing them with zero.

10. Tools, Files and Links

Item	Details
Source workbook	PMO Portfolio_From Strategy to Success_C41.xlsx (provided case-study workbook)
Exploration	Excel
Preparation and data quality	Power Query
Model, measures and dashboard	Power BI Desktop and DAX
GitHub / SQL scripts	[Add GitHub link — no SQL scripts completed for this version]
Power BI dashboard	[Add Power BI dashboard link or published report link]
Portfolio / README	[Add portfolio or README link]